Beach Day: A Full Sunrise-to-Sunset Coastal Scene in OpenGL

Course: CSE 4102 — Computer Graphics and Image Processing Laboratory

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Description

This project presents a calm 3D beach scene where a person sits under an umbrella and a small boat drifts on the sea, rendered across a full sunrise-to-sunset day cycle. The sun acts as a directional light source that moves across the sky, causing the shadows of the umbrella, palms, person, and boat to sweep and stretch throughout the day, while the light's color and intensity shift from warm sunrise tones to bright midday white and back to warm sunset hues. The scene demonstrates core transformation concepts (translation, rotation, scaling) through the moving sun, bobbing boat, and shifting tide line, combined with ambient, diffuse, and specular Phong lighting.

Main 3D Objects and Graphics Concepts

Graphics Concept	Applied To	Purpose in Scene
Translation	Sun, boat, tide (shoreline edge)	Positioning the sun along its arc, moving the boat across the sea, shifting the water's edge back and forth along the shore to simulate tide
Rotation	Sun, umbrella, palm leaves	Changing sun orientation for day cycle, tilting umbrella with wind, swaying palm leaves
Scaling	Shadows, sun disc	Stretching/shrinking shadows of umbrella, person, palms, and boat as the sun's angle changes; scaling the sun's disc size near the horizon
Perspective Projection	Overall scene camera	Realistic depth view of the full beach scene from the viewer's eye
Orthographic Projection	Optional top or side view	Alternate technical view for verifying object layout and scale
Hierarchical Modeling	Umbrella (pole to canopy)	Parent-child structure so tilting the pole automatically moves the canopy with it
Lighting and Object Coloring	Sun (light source), sand, sea, sky	Simulating sunlight color and intensity change, and realistic material coloring of scene objects
Flat Shading	Distant or simple objects (e.g., rocks)	Fast, faceted rendering for low-detail geometry
Gouraud Shading	Sea surface, sand terrain	Smooth color interpolation across curved or large surfaces
Phong Shading and Lighting	Person, umbrella, boat, palms	Realistic ambient, diffuse, and specular highlights on primary objects

Extra Features

- Skybox implementation for a realistic, seamless horizon and background
- Basic texture mapping on sand, water, boat sail, and palm leaves
- Day-cycle speed control (slider) to speed up or slow down the sunrise-to-sunset transition